

Assignment: Angular Core Concepts

◆ Section 1: One-Way & Two-Way Data Binding

✅ Conceptual Questions

1. Explain **one-way data binding** in Angular. List its types.
 2. What is the difference between:
 - Interpolation vs Property Binding?
 - Property Binding vs Event Binding?
 3. Explain **two-way data binding**. Why is it useful in forms?
 4. What is the role of `[(ngModel)]` in Angular?
 5. Can two-way binding work without importing any module? Justify your answer.
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Practical Tasks

Task 1: One-Way Binding Demo

Create a component:

- Display a student name using **interpolation**
 - Bind an image dynamically using **property binding**
 - Handle a button click using **event binding** to show an alert
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Task 2: Two-Way Binding Form

Create a simple form:

- Input fields: Name, Email
- Use `[(ngModel)]` for binding
- Display entered values live below the form

👉 Expected Output:

Name: John

Email: john@gmail.com

◆ Section 2: Angular Models

✓ Conceptual Questions

1. What is a **model** in Angular?
 2. Difference between:
 - Class vs Interface as a model
 3. Why do we use models in large applications?
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Practical Tasks

Task 3: Create a Model

Create a **Product model** with:

- id
 - name
 - price
 - quantity
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Task 4: Use Model in Component

- Create an array of products using the model
 - Display product data in UI
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◆ Section 3: ngIf and ngFor

✓ Conceptual Questions

1. What is a **structural directive**?
2. Difference between:
 - *ngIf vs hidden

3. What is *ngFor and how does it work internally?
 4. What is the use of trackBy in ngFor?
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Practical Tasks

Task 5: ngIf Example

- Create a login toggle
 - Show “Welcome User” only if logged in
 - Use a button to toggle login state
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Task 6: ngFor Example

- Display a list of courses:
 - Angular
 - React
 - .NET
 - Show index along with name
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Task 7: Combined Example

- Display products using ngFor
 - If product price > 500 → show "Premium Product"
 - Else → show "Budget Product"
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◆ Section 4: Angular Components

✔ Conceptual Questions

1. What is an Angular component?
2. Explain the role of:
 - @Component decorator

- selector
 - template
 - styles
3. What is the difference between:
- Inline template vs External template
4. What is component lifecycle (basic idea only)?
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Practical Tasks

Task 8: Create Basic Component

- Create a component named student
 - Display:
 - Name
 - Age
 - Course
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Task 9: Multiple Components

Create:

- Header Component
- Footer Component
- Dashboard Component

 Combine all in **App Component**

Task 10: Component Styling

- Apply different styles:
 - Inline styles
 - External CSS

- Highlight active section
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Mini Project (Final Assignment)

Build a Simple Product Dashboard

Requirements:

- Use a **Product model**
 - Display product list using ngFor
 - Use ngIf to show stock status:
 - In Stock / Out of Stock
 - Add input fields with **two-way binding**
 - Allow user to:
 - Add new product
 - Display it dynamically
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